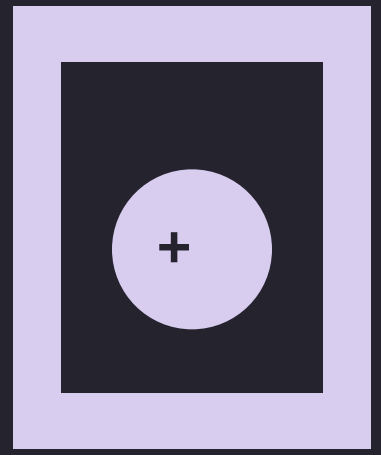
SecureVault

Private files.
Visible security controls.



An inspectable workflow for validation, reconstruction, encrypted storage and deliberate sharing.

[Student name] / [Enrollment number] / [Institution]

An upload is a trust decision.

The project makes that decision visible and testable.

1

Deceptive files

Names and MIME declarations can lie.

2

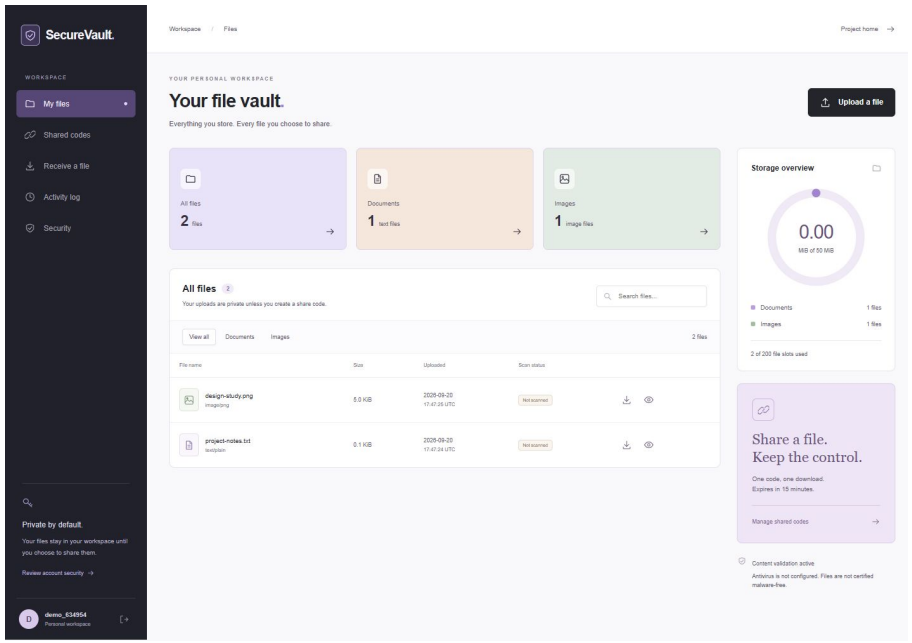
Unnecessary exposure

Metadata and public links can reveal information.

3

Missing evidence

Users need to know which checks actually ran.



Actual workspace capture with synthetic demo files

Research to implementation

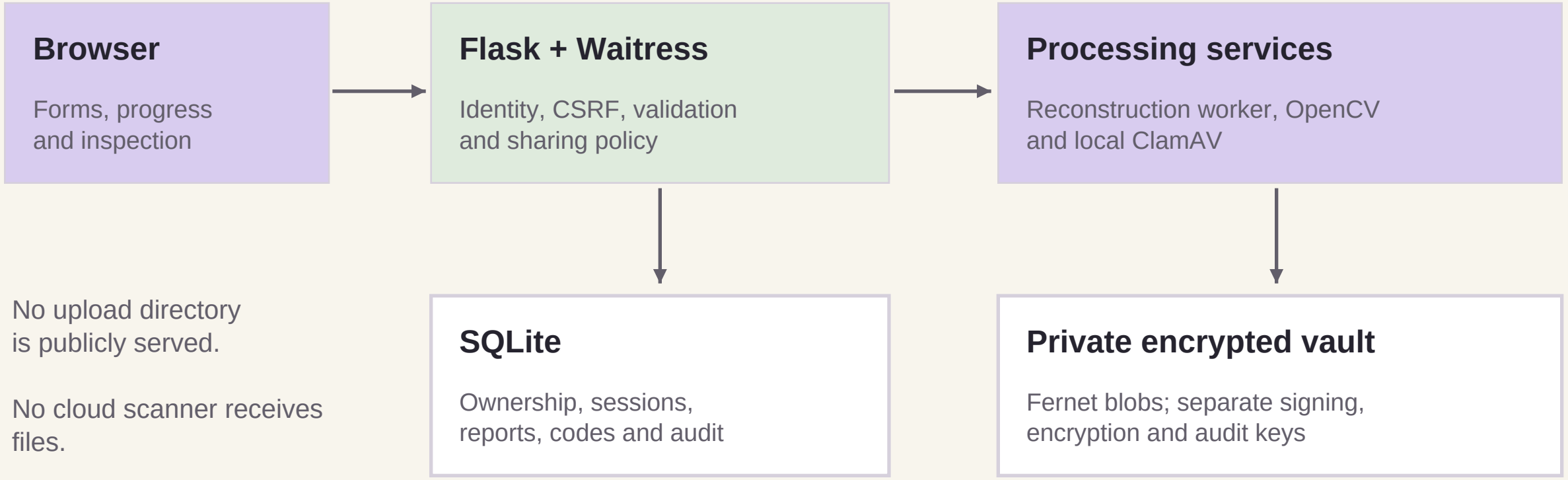
Three selected papers, translated into specific engineering choices.

Research	What it contributes	Our implementation
Eian et al. / 2020	Practical secure-development guidance	Security requirements become acceptance tests
Kudriavtseva & Gadyatskaya / 2022	Review of 28 secure SDLC methodologies	Design, operational responsibilities and evidence
Neef & Oudeh / 2024 / FUEL	Varied file-upload scanner evaluation	Negative cases; no reliance on antivirus alone

Design guidance: OWASP File Upload Cheat Sheet + NIST SP 800-218 SSDF

System architecture

Separate responsibilities. Explicit trust boundaries.



No upload directory is publicly served.

No cloud scanner receives files.

From upload to one-use delivery

The technical workflow—and where rejection happens.



Failure -> reject and record the reason. Sensitive-data findings -> private access only.

Technology you can locate in the code

A compact stack with clear ownership of each concern.

PROJECT MAP

```
app/
__init__.py
security_engines.py
content_worker.py
document_scanner.py
templates/ + static/
tests/ scripts/ docs/
```

Flask + SQLite + Waitress

Routes, transactions and sessions

Pillow + OpenCV + PDFium

Image handling and reconstruction

Fernet + scrypt + PyOTP

Encryption, passwords and MFA

pytest + Bandit + pip-audit

Regression and dependency evidence

Advanced features, honest outcomes

The interface separates document processing from malware checks.

Reconstruct

PDF -> image-only pages
DOCX -> plain paragraphs
CSV -> formula protection
HTML/XML/SVG -> restricted content

Inspect

ClamAV antivirus
YARA-X local rules
Tesseract OCR assisted DLP
Per-file report + SHA-256

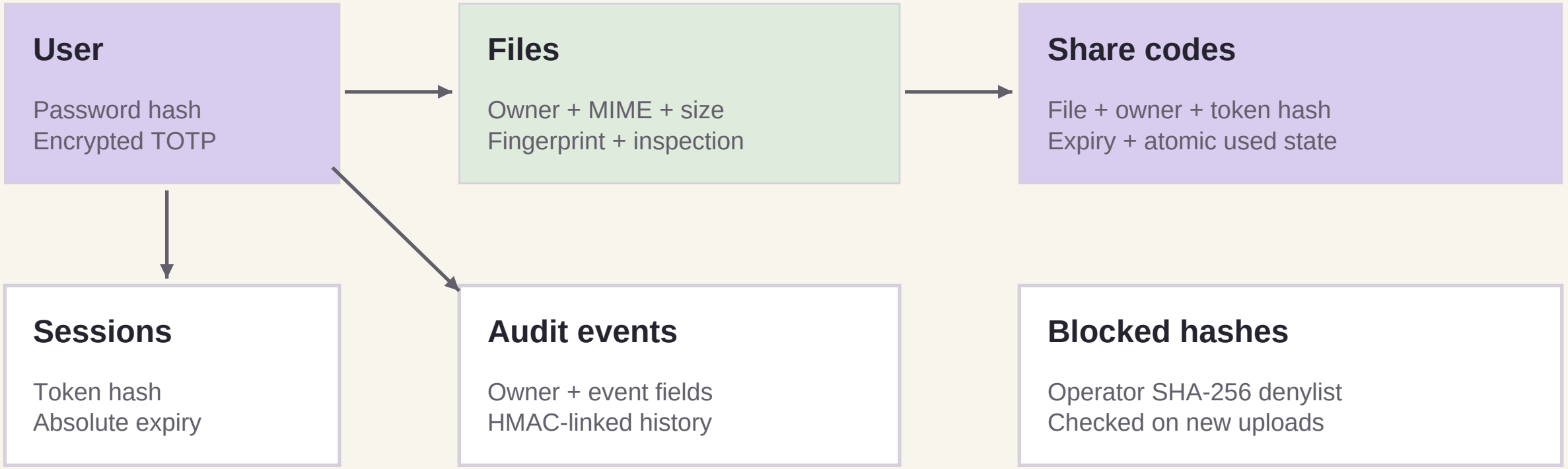
Control

OpenCV page enhancement
Private encrypted storage
DLP restriction on sharing
15-minute, single-use code

Supported: TXT · PNG · JPEG · PDF · DOCX · CSV · HTML · XML · restricted SVG

Access is a relationship, not a filename.

The data model makes ownership and sharing explicit.



Owner checks on every private route. File integrity is rechecked before delivery and reports.

Why this design is distinctive

The contribution is integrated, explainable security—not new cryptography.

Decision	Why it helps	Accepted tradeoff
Reconstruct instead of retain	Reduce original active document structure	Some fidelity is lost
One-use codes + DLP gate	Deliberate sharing with visible restrictions	Code delivery must be trusted
Per-file inspection evidence	Explain what ran and verify integrity now	Not an independent certificate

Explainable controls, visible outcomes and clear limits.

SDLC and future scope

Security requirements remain visible through delivery.

1 **Define** → 2 **Design** → 3 **Build** → 4 **Verify** → 5 **Operate**

Delivered

SRS + threat model + alternatives
Working local security controls
Regression and inspection evidence
Report, synopsis and 12-slide deck

Next

Licensed multi-engine / CDR integration
Hardened isolated sandbox service
Passkeys and managed key rotation
External audit checkpoints and scaling

Demonstrate. Explain. Defend.

Use synthetic samples and show the security report.

A five-step demonstration

01 Upload a supported sample.
02 Scan the document photo.
03 Inspect processing and AV results.
04 Share once; show reuse denial.
05 Explain DLP and integrity checks.

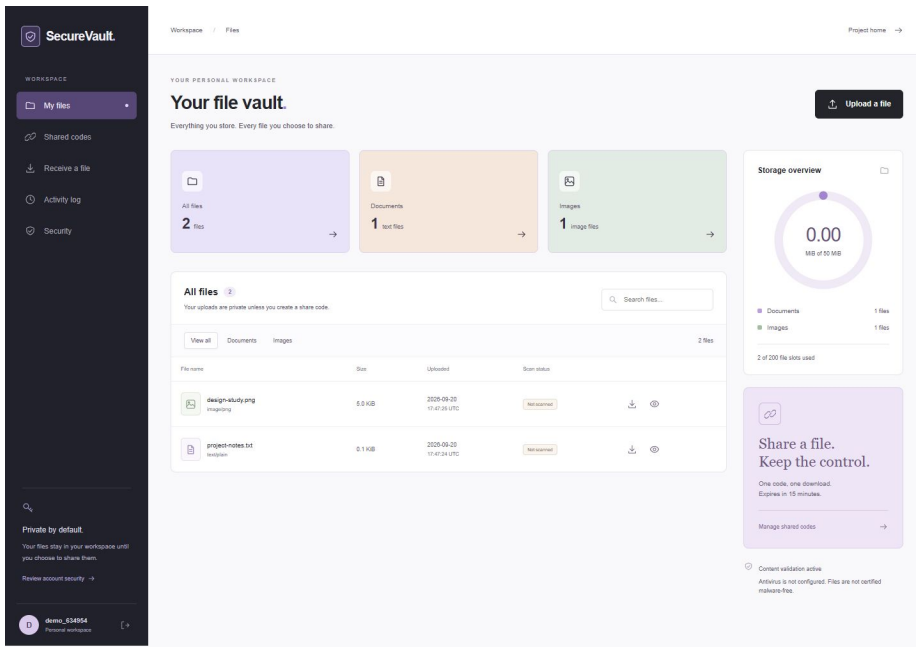
Primary references

OWASP / File Upload Cheat Sheet
NIST / SP 800-218 Secure SDLC
Eian (2020) / secure development
Kudriavtseva (2022) / methodology review
Neef (2024) / FUEL scanner evaluation
OpenCV, PDFium and ClamAV documentation

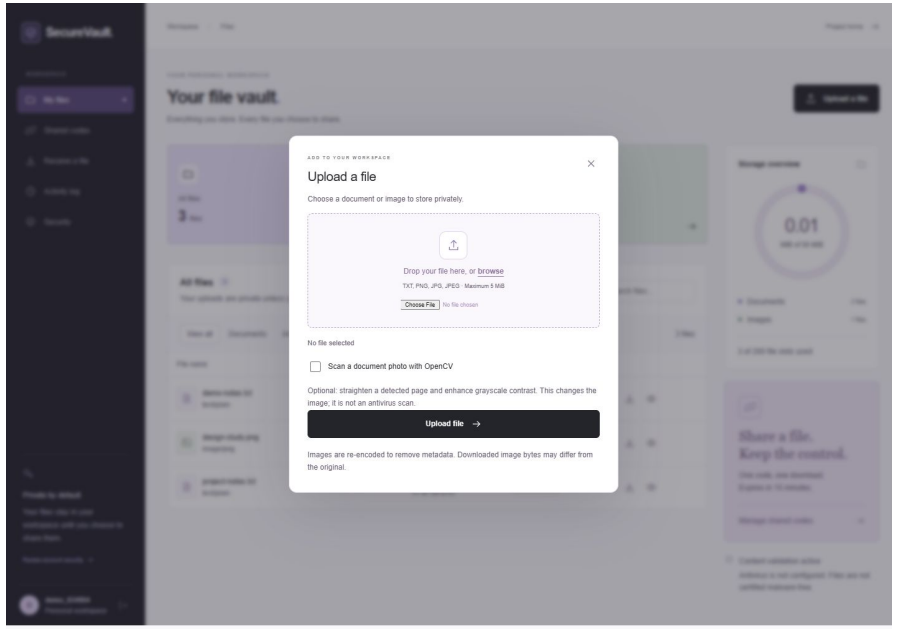
Full URLs: PROJECT_REPORT.md and RESEARCH.md

A visual tour of SecureVault

Workspace and upload screens from the project interface.



01 Private file workspace



02 Upload and OpenCV option

Interface captures are earlier project screenshots; show current scanner outcomes in the live inspection report.